

ALK fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ALK gene fusions from the msk_impact_50k_2026 study, covering 272 fusion events. 226/272 fusions (83.1%) were in-frame. 261/272 fusions (96.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.00191661$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for ALK.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 272 analyzed fusion events, identifying 32 distinct fusion partner genes. The most frequent partner was EML4, observed in 221 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions retained the domain, $p=0.00191661$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Domain disruption was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions disrupted the domain, $p=0.00768446$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.894511.

Cutpoint detection

Cutpoint detection scanned 271 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 972 aa (permutation-corrected $p=0.000999001$, statistically significant at $\alpha=0.05$). This is 12 aa from the nearest configured domain boundary at 960 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=261$) and "not_retained" ($n=10$). For Frame_status, retained: 222/227 (97.8%, 95% CI 94.9%-99.1%); not_retained: 4/5 (80.0%, 95% CI 37.6%-96.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 65.6139 vs 31.8, $p=0.00432792$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (32 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: ALK has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	221	0.8125	0.30004340774793187	0.004841453344022447	0.8865878486549185	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.5212516332281548	1
LRP1B	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	1.0	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2956163971063876	2
EMILIN1	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.9559412550066756	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28900758535738896	3
CLTC	2	0.007352941176470588	0.30004340774793187	0.004841453344022447	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28650572012531483	4
FN1	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2854027789488442	5
NOTCH1	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2854027789488442	6
NRP2	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2854027789488442	7
STRN	7	0.025735294117647058	0.30004340774793187	0.004841453344022447	0.8865153538050734	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28521134723597213	8
KIF5B	5	0.01838235294117647	0.30004340774793187	0.004841453344022447	0.8945260347129506	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28420706701921256	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PLEKHA7	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.9158878504672897	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28299957467648107	10
SPTBN1	4	0.014705882352941176	0.30004340774793187	0.004841453344022447	0.8895193591455274	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2823531245076285	11
RANBP2	3	0.011029411764705883	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28059931550739287	12
PICALM	3	0.011029411764705883	0.30004340774793187	0.004841453344022447	0.8811748998664887	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2799985144393021	13
SQSTM1	2	0.007352941176470588	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2794963743309223	14
TPM1	2	0.007352941176470588	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2794963743309223	15
CTNNA1	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	16
DCTN1	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	17
DIAPH2	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	18
DYSF	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
GCC2	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	20
HMBX1	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	21
PLEKHH2	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	22
PPP1CB	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	23
TFG	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	24
TRAF3	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	25
TTC27	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	26
VCL	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	27
ZFPM2	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2783934331544517	28
H6PD	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.7503337783711616	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2581664638620618	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CTSE	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.39118825100133514	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20429463475658788	30
SOS1	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	None	0.6853431332412958	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.17131340836045603	31
ASAP2	1	0.003676470588235294	0.30004340774793187	0.004841453344022447	0.0	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.14561639710638763	32

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
223	0	261	1	9	0.0	0.03690036900369004	1.4329692908744058
516	0	261	2	8	0.0	0.001230012300123001	2.9100905455940684
785	0	261	3	7	0.0	3.658029145347215e-05	4.436752838604533
939	2	259	3	7	0.018018018018018018	0.0003515999093716071	3.4539512455419974
972	2	259	4	6	0.011583011583011582	1.3820564716971664e-05	4.8594742110534295
1023	10	251	4	6	0.05976095617529881	0.0007961806348620647	3.098988389769786
1028	11	250	4	6	0.066	0.0010657190672512412	2.9723572639982643
1032	12	249	4	6	0.07228915662650602	0.0013947454739763106	2.8555050392241372
1035	13	248	4	6	0.07862903225806452	0.0017901726849544938	2.747105073765603
1040	14	247	4	6	0.08502024291497975	0.002258991897970247	2.646085326703862
1042	16	245	4	6	0.09795918367346938	0.003444927806265043	2.4628198749665864
1045	18	243	4	6	0.11111111111111111	0.005008607520231548	2.300282998750876
1047	19	242	4	6	0.11776859504132231	0.005949364209574388	2.225529443519691
1051	20	241	4	6	0.12448132780082988	0.007005060123461933	2.154588132872754
1052	22	239	4	6	0.13807531380753138	0.00948736346530418	2.0228544610771384
1058	254	7	10	0	0.0	1.0	-0.0
1059	255	6	10	0	0.0	1.0	-0.0
1061	258	3	10	0	0.0	1.0	-0.0
1064	259	2	10	0	0.0	1.0	-0.0
1070	260	1	10	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

222	40
4	5

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00629	PF00629	271	9	0	262	0.0
auto_disruption_pf12810	PF12810	271	7	3	261	0.0016115098351940458

domain_retention tables

frame_domain_contingency_table

222	39
4	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	271	261	0	10	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
20	254	272	0.9338235294117647

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ASAP2	1
CLTC	2

Partner_gene	Event_count
CTNNA1	1
CTSE	1
DCTN1	1
DIAPH2	1
DYSF	1
EMILIN1	1
EML4	221
FN1	1
GCC2	1
H6PD	1
HMBOX1	1
KIF5B	5
LRP1B	1
NOTCH1	1
NRP2	1
PICALM	3
PLEKHA7	1
PLEKHH2	1
PPP1CB	1
RANBP2	3
SOS1	1
SPTBN1	4
SQSTM1	2
STRN	7
TFG	1
TPM1	2
TRAF3	1
TTC27	1
VCL	1
ZFPM2	1

Per-event results (results.tsv)

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ role	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost _do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0024 679-T01-IM 6-39	P-0024 679-T0 1-IM6		ALK- CTS E	CTS E	unk now n	five _pr ime	295435 94	7	516	True	lost	0.0	False	PF006 29	Prot ein k inas e do mai n; P F12 810		ALK (NM_004304) - CTSE(NM_001910) Rearrangement: t(1;2)(q32. 1;p23.3)(chr1:g.206318648 ; chr2:g.29543594) Antisense Fusion	NA
EVT-P-0020 387-T01-IM 6-40	P-0020 387-T0 1-IM6		ALK- DCT N1	DCT N1	in-fr ame	thre e_ _pr ime	294468 96	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		DCTN1 (NM_004082) - ALK (NM_004304) rearrangement: c.3196+23 8:DCTN1_c.3173-502:ALK del Protein Fusion: in frame {DCTN1:ALK}	NA
EVT-P-0006 776-T01-IM 5-44	P-0006 776-T0 1-IM5		ALK- DYS F	DYS F	unk now n	five _pr ime	294474 18	19	1058	True	lost	0.0	False	PF006 29; PF 12810	Prot ein k inas e do mai n		ALK (NM_004304) - DYSF (NM_001130987) rearrangement: c.3172+90 9:ALK_c.3402+7348:DYSF dup Antisense fusion	NA
EVT-P-0031 827-T02-IM 6-45	P-0031 827-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294466 46	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490- 639:EML4_c.3173-252:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028 287-T01-IM 6-46	P-0028 287-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294466 61	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.4 94-2623:EML4_c.3173-267 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009 301-T01-IM 5-47	P-0009 301-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294466 51	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.668-6693: EML4_c.3173-257:ALKinv Protein fusion: in frame {EML4:ALK}	NA
EVT-P-0009 301-T02-IM 6-48	P-0009 301-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294466 51	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.668-6 693:EML4_c.3173-257:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0009 339-T01-IM 5-49	P-0009 339-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294478 89	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009 339-T02-IM 5-50	P-0009 339-T0 2-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294478 89	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009 343-T01-IM 5-51	P-0009 343-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 21	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-1093:EML4_c.3173-127:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009 353-T01-IM 5-52	P-0009 353-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294464 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.667+2407:EML4_c.3173-71:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0031 406-T01-IM 6-53	P-0031 406-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294464 25	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668-2889:EML4_c.3173-31:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 170-T01-IM 6-54	P-0032 170-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294476 95	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1822:EML4_c.3172+632:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0010 041-T01-IM 5-55	P-0010 041-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294479 42	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fuses with ALK exons 20-29) : c.701-6977:EML4_c.3172+385:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0008 332-T01-IM 5-56	P-0008 332-T0 1-IM5		ALK- EML 4	EML 4	unk now n	five _pr ime	294469 12	20	1058	True	lost	0.0	False	PF006 29; PF 12810	Prot ein k inas e do mai n		EML4 (NM_019063) - ALK (NM_004304) rearrangement : c.1489+828:EML4_c.3173-518:ALKdup Antisense fusion	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0010 177-T01-IM 5-57	P-0010 177-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294473 58	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0007 705-T01-IM 5-58	P-0007 705-T0 1-IM5		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294474 17	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29): c.1594:EML4_c.3172+910:ALKinv Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0010 978-T01-IM 5-59	P-0010 978-T0 1-IM5		ALK- EML 4	EML 4	out- of-fr ame	thr_ee _pr ime	294464 14	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0010 978-T02-IM 5-60	P-0010 978-T0 2-IM5		ALK- EML 4	EML 4	out- of-fr ame	thr_ee _pr ime	294464 14	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0011 018-T01-IM 5-61	P-0011 018-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294483 14	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011 203-T01-IM 5-62	P-0011 203-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 05	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011 203-T02-IM 5-63	P-0011 203-T0 2-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 96	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6805:EML4_c.3173-702:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011 203-T03-IM 6-64	P-0011 203-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 55	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 5 fused to ALK exon 20): c.494-6851:EML4_c.3173-661:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0012 679-T01-IM 5-65	P-0012 679-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 66	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012 778-T01-IM 5-66	P-0012 778-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 42	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320:EML4_c.3172+885:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012 778-T02-IM 6-67	P-0012 778-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 42	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320:EML4_c.3172+885:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0007 211-T01-IM 5-68	P-0007 211-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 89	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165: EML4_c.3172+838:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0007 125-T01-IM 5-69	P-0007 125-T0 1-IM5		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294465 97	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328: EML4_c.3173-203:ALKInv Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0006 950-T02-IM 6-70	P-0006 950-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294473 84	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons20-29): c.1489+2302:EML4_c.3172+943:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0013 202-T01-IM 5-71	P-0013 202-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 78	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+2565: EML4_c.3173-784:ALKInv Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006 950-T01-IM 5-72	P-0006 950-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294473 84	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1489+2302:EML4_c.3172+943: ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013 453-T01-IM 5-73	P-0013 453-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294465 97	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+551:EML4_c.3173-203:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013 864-T01-IM 5-74	P-0013 864-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294482 09	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1489+658:EML4_c.3172+118: ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013 869-T01-IM 5-75	P-0013 869-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294467 12	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318: ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014 270-T01-IM 6-76	P-0014 270-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294466 79	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294:EML4_c.3173-285:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014 310-T01-IM 6-77	P-0014 310-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294475 44	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.667+6128:EML4_c.3172+783:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006 370-T01-IM 5-78	P-0006 370-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294481 24	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKinv Protein fusion: in frame (EML4-ALK)	NA

event_id	sampl_e_id	pa_tie_n_t_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0014 385-T01-IM 6-79	P-0014 385-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294476 76	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1890:EML4_c.3172+651:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006 313-T01-IM 5-80	P-0006 313-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294479 66	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) -ALK (NM_004304) Fusion (EML4 exons 1 to 18 fused with exons 20 to 29 of ALK) Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0030 590-T01-IM 6-81	P-0030 590-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294477 04	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.494-5628:EML4_c.3173-350:ALKInv. Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0000 474-T01-IM 3-82	P-0000 474-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 02	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0000 557-T01-IM 3-83	P-0000 557-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 43	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0000 732-T01-IM 3-84	P-0000 732-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294475 37	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015 197-T01-IM 6-85	P-0015 197-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294480 09	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6 fused to ALK exon 20) : c.668-2108:EML4_c.3172+318:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015 434-T02-IM 6-86	P-0015 434-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294481 97	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKInv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005 321-T01-IM 5-87	P-0005 321-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294482 50	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 13 fused with ALK exon 20): c.2068+156 :EML4_c.3173-693:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0029 971-T01-IM 6-88	P-0029 971-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294478 07	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -119:EML4_c.3172+520:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 921-T03-IM 6-89	P-0015 921-T0 3-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294468 09	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.6 67+516:EML4_c.3173-415: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T01-IM 6-90	P-0015 961-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294465 26	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6 fused to ALK exon 20): c.668-666:EML 4_c.3173-132:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T02-IM 6-91	P-0015 961-T0 2-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294465 26	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6 666:EML4_c.3173-132:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T03-IM 6-92	P-0015 961-T0 3-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294465 26	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6 666:EML4_c.3173-132:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 827-T01-IM 6-93	P-0031 827-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294467 00	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-673:EML4_c.3173-306:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027 119-T01-IM 6-94	P-0027 119-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294467 11	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused to ALK exons 20-29): c.1490- 837:EML4_c.3173-317:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029 750-T01-IM 6-95	P-0029 750-T0 1-IM6		ALK-EML 4	EML 4	in-frame	three_prime	294466 60	20	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_c.3173-266:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029 550-T01-IM 6-96	P-0029 550-T0 1-IM6		ALK-EML 4	EML 4	in-frame	three_prime	294478 63	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016 280-T01-IM 6-97	P-0016 280-T0 1-IM6		ALK-EML 4	EML 4	in-frame	three_prime	294479 05	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016 580-T01-IM 6-98	P-0016 580-T0 1-IM6		ALK-EML 4	EML 4	in-frame	three_prime	294475 79	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0005 305-T01-IM 5-99	P-0005 305-T0 1-IM5		ALK-EML 4	EML 4	in-frame	three_prime	294477 64	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757:EML4_c.3172+563:ALKinv Protein fusion: in frame {EML4:ALK}	NA
EVT-P-0016 960-T01-IM 6-100	P-0016 960-T0 1-IM6		ALK-EML 4	EML 4	in-frame	three_prime	294478 38	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017 592-T01-IM 5-101	P-0017 592-T0 1-IM5		ALK-EML 4	EML 4	in-frame	three_prime	294475 48	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29) : c.494-5963:EML4_c.3172+779:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_n_t_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0029 400-T01-IM 6-102	P-0029 400-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294476 68	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 315+641:EML4_c.3172+65 9:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0004 218-T01-IM 5-103	P-0004 218-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 87	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156 :EML4_c.3173-693:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0017 978-T01-IM 6-104	P-0017 978-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7 306:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017 978-T02-IM 6-105	P-0017 978-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7 306:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017 978-T03-IM 6-106	P-0017 978-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7 306:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018 242-T01-IM 6-107	P-0018 242-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294480 07	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused in-frame with ALK exons 20-29): c.668-7560:EML4_ c.3172+320:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018 280-T01-IM 6-108	P-0018 280-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 87	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 6 fused with ALK exon 20): c.494-6792: EML4_c.3173-793:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0003 913-T01-IM 3-109	P-0003 913-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294472 11	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0029 174-T01-IM 6-110	P-0029 174-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 56	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-19 fused with ALK exons 20-29): c.2 242+168:EML4_c.3173-26 2:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018 969-T01-IM 6-112	P-0018 969-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294484 16	19	1028	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-760:EML4_c.3083:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019 491-T01-IM 6-113	P-0019 491-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 96	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1490- 486:EML4_c.3173-202:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019 513-T01-IM 6-114	P-0019 513-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294484 03	19	1032	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1 489+2278:EML4_c.3096:A LKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019 102-T01-IM 6-115	P-0019 533-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294464 94	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837: EML4_c.3173-100:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009 102-T01-IM 5-116	P-0009 102-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 13	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.149 0-244:EML4_c.3173-119:A LKinv Protein fusion: in frame (EML4-ALK)	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ role	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0028 901-T01-IM 6-117	P-0028 901-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294466 07	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 316-844:EML4_c.3173-213 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019 786-T01-IM 6-118	P-0019 786-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294475 75	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c .3172+752:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019 972-T01-IM 6-119	P-0019 972-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294481 01	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1 179:EML4_c.3172+226:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 124-T01-IM 6-120	P-0020 124-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294479 55	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_ c.3172+372:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 138-T01-IM 6-121	P-0020 138-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294479 07	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-3 016:EML4_c.3172+420:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003 383-T01-IM 5-122	P-0003 383-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294475 72	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0020 408-T01-IM 6-123	P-0020 408-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294467 79	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+ 1227:EML4_c.3173-385:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0003 323-T01-IM 5-124	P-0003 323-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294475 30	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003 297-T01-IM 5-125	P-0003 297-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294482 59	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0020 584-T01-IM 6-126	P-0020 584-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 48	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exon 20-29) : c.209-48 46:EML4_c.3173-654:ALKi nv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 780-T01-IM 6-127	P-0020 780-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 84	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1 - 13 fused to ALK exons 20 - 29): c.1 489+1278:EML4_c.3173-6 90:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 825-T03-IM 6-128	P-0020 825-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294467 67	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29):c.208+4 890:EML4_c.3173-373:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0021 257-T01-IM 6-129	P-0021 257-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294477 77	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-17 fused with ALK exons 20-29): c.1 883-596:EML4_c.3172+55 0:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003 206-T01-IM 5-130	P-0003 206-T0 1-IM5		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294483 65	19	1045	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0021 495-T01-IM 6-131	P-0021 495-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294483 13	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.4 93+2355:EML4_c.3172+14 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0002 613-T01-IM 3-132	P-0002 613-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 99	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0002 408-T02-IM 5-133	P-0002 408-T0 2-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294480 53	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29) ; c.2 057-115:EML4_c.3172+27 4:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0021 693-T01-IM 6-134	P-0021 693-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 02	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-14 fused with ALK exons 20-29): c.1 316-1509:EML4_c.3173-60 8:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 101-T01-IM 6-135	P-0022 101-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294477 59	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon18 fused with ALK exon 20): c.2057-510: EML4_c.2057-510:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 138-T01-IM 6-136	P-0022 138-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294465 84	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2 333:EML4_c.3173-190:AL Kinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0036 247-T01-IM 6-137	P-0036 247-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294478 57	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+107:EML4_c.3172+470: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036 106-T01-IM 6-138	P-0036 106-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294469 30	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490- 2055:EML4_c.3173-536:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028 745-T01-IM 6-139	P-0028 745-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294475 09	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 523-2563:EML4_c.3172:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0022 398-T01-IM 6-140	P-0022 398-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294480 35	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.4 94-7853:EML4_c.3172+29 2:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002 266-T01-IM 3-141	P-0002 266-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	five _pr ime	294472 88	20	1058	True	lost	0.0	False	PF006 29; PF 12810	Prot ein k inas e do mai n		ALK (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173-889: ALK_c.208+1022:EML4inv Protein fusion: in frame (ALK-EML4)	NA
EVT-P-0002 266-T01-IM 3-142	P-0002 266-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294472 83	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028 541-T03-IM 6-143	P-0028 541-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294469 22	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion: c.208+4631:EML4_c.3173- 528:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 926-T01-IM 6-144	P-0022 926-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 58	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.4 93+983:EML4_c.3173-164: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0035 121-T01-IM 6-145	P-0035 121-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294473 83	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -4857:EML4_c.3172+944:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002 217-T02-IM 3-146	P-0002 217-T0 2-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 94	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0023 368-T01-IM 6-147	P-0023 368-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294467 13	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain) : c.667+7004:EML 4_c.3173-319:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0002 024-T01-IM 3-148	P-0002 024-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 33	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) inversion: c. 2242+31:EML4_c.3173-73 9:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0023 962-T01-IM 6-149	P-0023 962-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294488 20	19	1023	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.6 67+4775:EML4_c.3068-38 9:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0024 136-T01-IM 6-150	P-0024 136-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294482 66	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4 498:EML4_c.3172+61:ALKI nv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0001 811-T01-IM 3-151	P-0001 811-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294482 05	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666:EM L4_c.3172+122:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028 541-T02-IM 6-152	P-0028 541-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294469 22	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.2 08+4631:EML4_c.3173-52 8:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0024 907-T01-IM 6-153	P-0024 907-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294468 77	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209-9 90:EML4_c.3173-483:ALKI nv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034 569-T02-IM 6-154	P-0034 569-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 14	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243- 217:EML4_c.3173-120:AL KInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034 450-T01-IM 6-155	P-0034 450-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294463 54	20	1071	False	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK exons 20) : c.1641+55 5:EML4_c.3213:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0001 532-T01-IM 3-156	P-0001 532-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294475 53	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028 541-T01-IM 6-157	P-0028 541-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294469 22	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.2 08+4631:EML4_c.3173-52 8:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0026 407-T01-IM 6-158	P-0026 407-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294463 92	20	1059	False	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1 981-2450:EML4_c.3175:AL Kinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0026 650-T02-IM 6-159	P-0026 650-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	five _pr ime	294474 51	19	1058	True	lost	0.0	False	PF006 29; PF 12810	Prot ein k inas e do mai n		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.6 67+7374:EML4_c.3172+87 6:ALKinv Protein Fusion: in frame {ALK:EML4}	NA
EVT-P-0026 657-T01-IM 6-160	P-0026 657-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294469 52	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2 312:EML4_c.3173-558:AL Kinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0026 657-T03-IM 6-161	P-0026 657-T0 3-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294469 52	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312: EML4_c.3173-558:ALKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0015 152-T02-IM 6-162	P-0015 152-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 79	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused in-frame to ALK exons 20-29): c.2242+127:EML4_ c.3173-685:ALKinv Protein fusion: in frame (EML4-ALK)	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0032 862-T01-IM 6-163	P-0032 862-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre_e _pr ime	294474 88	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) Fusion(EML4 exon 11 fused to ALK exon20): c.1489+671:EML4_c.3172+839:ALK Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027 119-T03-IM 6-164	P-0027 119-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thre_e _pr ime	294467 11	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-837:EML4_c.3173-317:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 683-T01-IM 6-165	P-0032 683-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre_e _pr ime	294477 63	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028 339-T01-IM 6-166	P-0028 339-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre_e _pr ime	294476 80	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+1245:EML4_c.3172+647:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 518-T01-IM 6-167	P-0032 518-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thre_e _pr ime	294483 59	19	1047	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+2607:EML4_c.3140:ALKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0027 853-T01-IM 6-168	P-0027 853-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre_e _pr ime	294477 34	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-157:EML4_c.3172+593:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 110-T01-IM 6-169	P-0015 110-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre_e _pr ime	294465 18	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668-1726: EML4_c.3173-124:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003 446-T01-IM 5-170	P-0003 446-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre_e _pr ime	294465 34	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		NA Protein fusion: in frame (EML4-ALK)	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0004 516-T02-IM 5-171	P-0004 516-T0 2-IM5		ALK- GCC 2	GC C2	in-fr ame	thr_ee _pr ime	294465 26	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		GCC2 (NM_181453) - ALK (NM_004304) rearrangement : c.4448-12 30:GCC2_c.3173-132:ALKI nv Protein fusion: in frame (GCC2-ALK)	NA
EVT-P-0035 712-T01-IM 6-172	P-0035 712-T0 1-IM6		ALK- H6P D	H6P D	unk now n	five _pr ime	294625 08	13	785	True	lost	0.0	False	PF006 29	Prot ein k inas e do mai n	PF1281 0	ALK (NM_004304) rearrangement: t(1;2)(p36. 22;p23.2)(chr1:g.9314150:: chr2:g.29462508) Antisense Fusion	NA
EVT-P-0018 541-T01-IM 6-174	P-0018 541-T0 1-IM6		ALK- LRP 1B	LRP 1B	in-fr ame	five _pr ime	294502 89	17	972	True	lost	0.0	False	PF006 29; PF 12810	Prot ein k inas e do mai n		ALK (NM_004304) - LRP1B (NM_018557) Rearrangement : c.2914+1 51:ALK_c.344-85984:LRP1 Bdup Protein Fusion: in frame {ALK:LRP1B}	NA
EVT-P-0027 092-T01-IM 6-176	P-0027 092-T0 1-IM6		ALK- PLE KHH 2	PLE KHH 2	in-fr ame	thr_ee _pr ime	294465 06	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		PLEKHH2 (NM_172069) - ALK (NM_004304) rearrangement: c.502+545: PLEKHH2_c.3173-112:AL Kinv Protein Fusion: in frame {PLEKHH2:ALK}	NA
EVT-P-0027 363-T02-IM 6-177	P-0027 363-T0 2-IM6		ALK- RAN BP2	RAN BP2	in-fr ame	thr_ee _pr ime	294473 64	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-155 6:RANBP2_c.3172+963:AL Kinv Protein Fusion: in frame {RANBP2:ALK}	NA
EVT-P-0068 865-T02-IM 7-178	P-0068 865-T0 2-IM7		ALK- RAN BP2	RAN BP2	in-fr ame	five _pr ime	294473 73	19	1058	True	lost	0.0	False	PF006 29; PF 12810	Prot ein k inas e do mai n		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.3172+95 4:ALK_c.2602+1532:RANB P2inv Protein Fusion: in frame {ALK:RANBP2}	NA
EVT-P-0008 843-T01-IM 5-179	P-0008 843-T0 1-IM5		ALK- RAN BP2	RAN BP2	in-fr ame	thr_ee _pr ime	294464 88	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-107 0:RANBP2_c.3173-94:ALKI nv Protein fusion: in frame (RANBP2-ALK)	NA
EVT-P-0015 699-T01-IM 6-181	P-0015 699-T0 1-IM6		ALK- SPT BN1	SPT BN1	in-fr ame	thr_ee _pr ime	294474 37	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion (SPTBN1 exon 7 fused to ALK exon 20): c.764-1028: SPTBN1_c.3172+890:ALKI nv Protein fusion: in frame (SPTBN1-ALK)	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ role	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0016 193-T05-IM 6-182	P-0016 193-T0 5-IM6		ALK- STR N	STR N	unk now n	thre e_ _pr ime	294483 26	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.4 12+4919:STRN_c.3172+1: ALKdel Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0016 193-T02-IM 6-183	P-0016 193-T0 2-IM6		ALK- STR N	STR N	unk now n	thre e_ _pr ime	294483 26	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		ALK (NM_004304) - STRN (NM_003162) :c.3172+1:A LK_c.412+4919:STRNdel Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0016 193-T01-IM 6-184	P-0016 193-T0 1-IM6		ALK- STR N	STR N	unk now n	thre e_ _pr ime	294483 26	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		ALK (NM_004304) - STRN (NM_003162) :c.3172+1:A LK_c.412+4919:STRNdel Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0041 438-T01-IM 6-185	P-0041 438-T0 1-IM6		ALK- TTC 27	TTC 27	out- of-fr ame	five _pr ime	294478 17	19	1058	True	lost	0.0	False	PF006 29; PF 12810	Prot ein k inas e do mai n		ALK (NM_004304) - TTC27 (NM_017735) rearrangement: c.3172+51 0:ALK_c.538-4165:TTC27i nv Protein Fusion: out of frame {ALK:TTC27}	NA
EVT-P-0006 617-T01-IM 5-187	P-0006 617-T0 1-IM5		ASA P2-A LK	ASA P2	unk now n	five _pr ime	301427 53	1	223	True	lost	0.0	False		Prot ein k inas e do mai n; P F00 629; PF1 281 0		ASAP2 (NM_003887) - ALK (NM_004304) rearrangement event : c.12 7-20148:ASAP2_c.667+10 6:ALKdel Antisense fusion	NA
EVT-P-0062 994-T02-IM 7-188	P-0062 994-T0 2-IM7		CLT C-AL K	CLT C	in-fr ame	thre e_ _pr ime	294496 24	18	1023	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17) (p23.2;q23.1)(chr2:g.2944 9624::chr17:g.57768350) Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0051 864-T01-IM 6-189	P-0051 864-T0 1-IM6		CLT C-AL K	CLT C	in-fr ame	thre e_ _pr ime	294488 39	19	1023	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17) (p23.2;q23.1)(chr2:g.2944 8839::chr17:g.57770376) Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0025 388-T01-IM 6-190	P-0025 388-T0 1-IM6		CTN NA1- ALK	CTN NA1	in-fr ame	thre e_ _pr ime	294464 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		CTNNA1 (NM_001903) - ALK (NM_004304) rearrangement: t(2;5)(p23.2; q31.2)(chr2:g.29446465::c hr5:g.138259019) Protein Fusion: in frame {CTNNA1:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0051 068-T01-IM 6-191	P-0051 068-T0 1-IM6		DIAP H2-A LK	DIA PH2	in-fr ame	thr ee _pr ime	294475 66	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		DIAPH2 (NM_006729) - ALK (NM_004304) fusion: t (2;X)(p23.2;q21.33)(chr2:g. 29447566::chrX:g.9671362 0) Protein Fusion: in frame {DIAPH2:ALK}	NA
EVT-P-0005 759-T02-IM 6-192	P-0005 759-T0 2-IM6		EMIL IN1- ALK	EMI LIN1	in-fr ame	thr ee _pr ime	294507 41	17	939	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9	PF1281 0	EMILIN1 (NM_007046) - ALK (NM_004304) fusion: c.2575+44:EMILIN1_c.281 6-203:ALKinv Protein Fusion: in frame {EMILIN1:ALK}	NA
EVT-P-0046 083-T01-IM 6-193	P-0046 083-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294466 65	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+986:EML4_c.3173-271:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049 375-T01-IM 6-194	P-0049 375-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294467 32	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +1166:EML4_c.3173-338:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 646-T03-IM 6-195	P-0015 646-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294480 28	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.209 -3909:EML4_c.3172+299:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014 385-T03-IM 6-196	P-0014 385-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 76	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1890:EML4_c.3172+651 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027 119-T04-IM 7-197	P-0027 119-T0 4-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294467 11	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-837:EML4_c.3173-317:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0057 391-T02-IM 6-198	P-0057 391-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 92	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -3018:EML4_c.3173-698:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034 334-T01-IM 6-199	P-0034 334-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 70	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4909:EML4_c.3173-176:A LKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0025 095-T01-IM 6-200	P-0025 095-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294473 21	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0025 040-T01-IM 6-201	P-0025 040-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294471 52	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20) : c.1316-1940:EML4_c.3173-758:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0007 705-T02-IM 6-202	P-0007 705-T0 2-IM6		EML 4-AL K	EML 4	unk now n	thr_ee _pr ime	294474 17	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0023 375-T02-IM 7-203	P-0023 375-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294478 54	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2728:EML4_c.3172+473:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0035 085-T03-IM 6-204	P-0035 085-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294470 79	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7361:EML4_c.3173-685:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0001 766-T04-IM 6-205	P-0001 766-T0 4-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294487 67	19	1023	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3962:EML4_c.3068-336:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0068 467-T02-IM 7-206	P-0068 467-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294482 95	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1566:EML4_c.3172+32:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 184-T04-IM 6-207	P-0022 184-T0 4-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294470 35	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1386:EML4_c.3173-641:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036 106-T03-IM 7-208	P-0036 106-T0 3-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294469 30	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-536:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie nt_id	fusio n_name	part ner_gene	fra me_status	tar get_role	breakp oint_ge nomic	break point_exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost _do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0022 184-T01-IM 6-209	P-0022 184-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294470 35	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 315+1386:EML4_c.3173-6 41:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 404-T01-IM 6-210	P-0037 404-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294478 72	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4963:EML4_c.3172+455: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 491-T01-IM 6-211	P-0037 491-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294467 86	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon12 fused with ALK exon20): c.1490-1120: EML4_c.3173-392:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 668-T01-IM 6-212	P-0037 668-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294476 31	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2742:EML4_c.3172+696: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 925-T01-IM 6-213	P-0037 925-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294465 22	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4634:EML4_c.3173-128:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 982-T01-IM 6-214	P-0037 982-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294464 79	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-138:EML4_c.3173-85:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038 201-T03-IM 6-215	P-0038 201-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294477 48	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+989:EML4_c.3172+579: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038 772-T01-IM 6-216	P-0038 772-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294466 87	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.234 1+147:EML4_c.3173-293:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 860-T02-IM 6-217	P-0043 860-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294469 31	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-980:EML4_c.3173-537:A LKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0067 093-T02-IM 7-218	P-0067 093-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294473 03	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +1508:EML4_c.3173-909:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017 592-T04-IM 6-219	P-0017 592-T0 4-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294475 48	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -5963:EML4_c.3172+779:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 261-T01-IM 6-220	P-0040 261-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294471 03	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1340:EML4_c.3173-709: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T07-IM 6-221	P-0015 961-T0 7-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 26	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -6666:EML4_c.3173-132:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 553-T01-IM 6-222	P-0040 553-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 55	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+30:EML4_c.3173-661:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 574-T01-IM 6-223	P-0040 574-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294477 38	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-1589: EML4_c.3172+589:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 576-T01-IM 6-224	P-0040 576-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294473 21	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-821:E ML4_c.3173-927:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 600-T01-IM 6-225	P-0040 600-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 88	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +1795:EML4_c.3172+639: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 683-T02-IM 7-226	P-0032 683-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294477 63	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2122:EML4_c.3172+564: ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_n_t_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0014 385-T06-IM 7-227	P-0014 385-T0 6-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 76	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1890:EML4_c.3172+651 :ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067 069-T01-IM 7-228	P-0067 069-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294481 69	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+2305:EML4_c.3172+158 :ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014 385-T02-IM 6-229	P-0014 385-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 76	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1890:EML4_c.3172+651 :ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041 967-T01-IM 6-230	P-0041 967-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294481 47	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+447:EML4_c.3172+180: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041 967-T02-IM 7-231	P-0041 967-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294481 47	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+447:EML4_c.3172+180: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067 048-T02-IM 7-232	P-0067 048-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294473 91	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +2216:EML4_c.3172+936: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0066 568-T02-IM 7-233	P-0066 568-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294480 72	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +3428:EML4_c.3172+255: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014 310-T02-IM 6-234	P-0014 310-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294475 44	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +6128:EML4_c.3172+783: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 041-T01-IM 6-235	P-0043 041-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 97	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1782:EML4_c.3173-203: ALKInv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0013 299-T02-IM 6-236	P-0013 299-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 38	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) -ALK (NM_004304) Fusion(EML4 exon5 fused with ALK exon20) : c.667+ 1302:EML4_c.3173-644:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012 913-T02-IM 6-237	P-0012 913-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294469 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.209 -4133:EML4_c.3173-571:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 350-T01-IM 6-238	P-0043 350-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294475 13	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -4017:EML4_c.3172+814:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 363-T01-IM 6-239	P-0043 363-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294481 01	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.225 4:EML4_c.3172+226:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0043 594-T01-IM 6-240	P-0043 594-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294483 81	19	1040	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.339 -2119:EML4_c.3118:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0069 396-T02-IM 7-241	P-0069 396-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294469 44	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -4308:EML4_c.3173-550:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069 535-T01-IM 7-242	P-0069 535-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294477 64	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +3829:EML4_c.3172+563: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069 617-T01-IM 7-243	P-0069 617-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294480 17	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +7821:EML4_c.3172+310: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044 432-T01-IM 6-244	P-0044 432-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294483 44	19	1052	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.205 7-576:EML4_c.3155:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0005 743-T02-IM 5-245	P-0005 743-T0 2-IM5		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 69	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243- 181:EML4_c.3172+658:AL Kinv Protein fusion: in frame {EML4-ALK}	NA
EVT-P-0045 222-T03-IM 7-246	P-0045 222-T0 3-IM7		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294475 83	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.227 7:EML4_c.3172+744:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0045 243-T01-IM 6-247	P-0045 243-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294482 98	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -3064:EML4_c.3172+29:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0045 341-T01-IM 6-248	P-0045 341-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294479 20	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -4434:EML4_c.3172+407:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065 469-T02-IM 7-249	P-0065 469-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -1949:EML4_c.3173-171:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065 210-T01-IM 7-250	P-0065 210-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294480 76	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -2012:EML4_c.3172+251:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012 873-T03-IM 7-251	P-0012 873-T0 3-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294464 39	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+2231:EML4_c.3173-45:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064 716-T02-IM 7-252	P-0064 716-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294477 39	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -3669:EML4_c.3172+588:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 368-T04-IM 6-253	P-0031 368-T0 4-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294471 91	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1154:EML4_c.3173-797: ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0045 912-T02-IM 6-254	P-0045 912-T0 2-IM6		EML 4-AL K	EML 4	out-of-fr ame	thr_ee _pr ime	294465 07	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.215 4+1011:EML4_c.3173-113: ALKinv Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0031 824-T01-IM 6-255	P-0031 824-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294505 57	17	939	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9	PF1281 0	EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +2971:EML4_c.2816-19:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046 083-T02-IM 7-256	P-0046 083-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294466 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+986:EML4_c.3173-271:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046 594-T01-IM 6-257	P-0046 594-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294464 45	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +2217:EML4_c.3173-51:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 732-T02-IM 6-258	P-0040 732-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294466 36	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1832:EML4_c.3173-242: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0047 355-T01-IM 6-259	P-0047 355-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +925:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0047 355-T02-IM 6-260	P-0047 355-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +925:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064 342-T01-IM 7-261	P-0064 342-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294465 34	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +4042:EML4_c.3173-140:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0063 585-T01-IM 7-262	P-0063 585-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294468 84	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -2813:EML4_c.3173-490:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012 873-T02-IM 6-263	P-0012 873-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294464 39	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+2231:EML4_c.3173-45:A LKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0062 999-T02-IM 7-264	P-0062 999-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294470 91	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2055:EML4_c.3173-697: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049 318-T01-IM 6-265	P-0049 318-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr_ee _pr ime	294463 59	20	1070	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.215 4+2335:EML4_c.3208:ALKI nv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0046 720-T01-IM 6-266	P-0046 720-T0 1-IM6		EML 4-AL K	EML 4	out- of-fr ame	thr_ee _pr ime	294466 69	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.164 1+1:EML4_c.3173-275:AL Kinv Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0049 375-T02-IM 6-267	P-0049 375-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294467 32	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +1166:EML4_c.3173-338:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049 511-T01-IM 6-268	P-0049 511-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294469 91	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.224 2+9:EML4_c.3173-597:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0010 177-T02-IM 7-269	P-0010 177-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294473 58	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2649:EML4_c.3173-964: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 827-T03-IM 6-270	P-0031 827-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294466 46	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-639:EML4_c.3173-252:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051 231-T01-IM 6-271	P-0051 231-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294467 74	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4130:EML4_c.3173-380:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051 231-T03-IM 6-272	P-0051 231-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294467 74	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4130:EML4_c.3173-380:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0062 985-T02-IM 7-273	P-0062 985-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294471 99	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4561:EML4_c.3173-805:A LKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ role	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0052 035-T03-IM 6-274	P-0052 035-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294476 56	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.224 2+79:EML4_c.3172+671:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0061 297-T03-IM 7-275	P-0061 297-T0 3-IM7		EML 4-AL K	EML 4	unk now n	thre e_ _pr ime	294483 75	19	1042	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+931:EML4_c.3124:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0052 297-T02-IM 6-276	P-0052 297-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294482 55	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-161:EML4_c.3172+72:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0061 297-T02-IM 7-277	P-0061 297-T0 2-IM7		EML 4-AL K	EML 4	unk now n	thre e_ _pr ime	294483 75	19	1042	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+931:EML4_c.3124:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0005 786-T01-IM 5-278	P-0005 786-T0 1-IM5		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294482 39	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 with ALK exons 20-29) : c.2057- 186:EML4_c.3172+88:ALKi nv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0060 148-T01-IM 7-279	P-0060 148-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294477 57	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+2600:EML4_c.3172+570 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0059 699-T03-IM 7-280	P-0059 699-T0 3-IM7		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294474 67	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1990:EML4_c.3172+860: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 827-T06-IM 7-281	P-0031 827-T0 6-IM7		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294466 46	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-639:EML4_c.3173-252:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054 263-T01-IM 6-282	P-0054 263-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294465 57	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -210:EML4_c.3173-163:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0059 257-T02-IM 7-283	P-0059 257-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294482 10	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -982:EML4_c.3172+117:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054 408-T01-IM 6-284	P-0054 408-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294469 02	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.224 3-262:EML4_c.3173-508:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0008 332-T02-IM 6-285	P-0008 332-T0 2-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294469 13	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828:EML4_c.3173- 519:ALKdup Antisense Fusion	NA
EVT-P-0058 599-T02-IM 7-286	P-0058 599-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294464 15	20	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1345:EML4_c.3173-21:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044 042-T01-IM 6-287	P-0044 042-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294477 27	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -1681:EML4_c.3172+600:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019 560-T01-IM 6-288	P-0019 560-T0 1-IM6		EML 4-AL K	EML 4	out- of-fr ame	thr ee _pr ime	294479 52	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-16 with ALK exons 20-29): c.1968- 2879:EML4_c.3172+375:A LKinv Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0044 526-T02-IM 6-289	P-0044 526-T0 2-IM6		FN1- ALK	FN1	in-fr ame	thr ee _pr ime	294485 33	19	1023	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		FN1 (NM_212482) - ALK (NM_004304) fusion: c.181 9+332:FN1_c.3068-102:AL Kdel Protein Fusion: in frame {FN1:ALK}	NA
EVT-P-0030 808-T01-IM 6-290	P-0030 808-T0 1-IM6		HMB OX1- ALK	HM BOX 1	unk now n	thr ee _pr ime	294477 13	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		HMBBOX1 (NM_001135726) ALK (NM_004304) Rearrangement : t(2;8)(p23 .3,p21.1)(chr2:g.29447713: :chr8:g.28904937) Protein Fusion: mid-exon {HMBBOX1:ALK}	NA
EVT-P-0062 804-T03-IM 7-291	P-0062 804-T0 3-IM7		KIF5 B-AL K	KIF5 B	in-fr ame	thr ee _pr ime	294477 91	19	1058	True	retain ed	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)p23.2;p11.22)(chr2:g.294 47791::chr10:g.32304833) Protein Fusion: in frame {KIF5B:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005 711-T02-IM 6-292	P-0005 711-T0 2-IM6		KIF5 B-ALK	KIF5 B	in-frame	threep _prime	294495 48	18	1023	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29): t(2;10)(p23.2;p11.22)(chr2:g.29449548::chr10:g.32304765) Protein fusion: in frame (KIF5B-ALK)	NA
EVT-P-0008 764-T01-IM 5-293	P-0008 764-T0 1-IM5		KIF5 B-ALK	KIF5 B	unknown	threep _prime	294483 27	19	1058	False	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29): t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750) Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0017 924-T01-IM 6-294	P-0017 924-T0 1-IM6		KIF5 B-ALK	KIF5 B	in-frame	threep _prime	294479 45	19	1058	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447945::chr10:g.32305310) Protein Fusion: in frame {KIF5B:ALK}	NA
EVT-P-0008 764-T02-IM 6-295	P-0008 764-T0 2-IM6		KIF5 B-ALK	KIF5 B	unknown	threep _prime	294483 27	19	1058	False	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29): t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750) Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0067 512-T01-IM 7-297	P-0067 512-T0 1-IM7		NOTCH1- ALK	NOTCH1	in-frame	threep _prime	294491 52	18	1023	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		NOTCH1 (NM_017617) - ALK (NM_004304) fusion: t(2;9)(p23.2;q34.3)(chr2:g.29449152::chr9:g.139416748) Protein Fusion: in frame {NOTCH1:ALK}	NA
EVT-P-0034 044-T01-IM 6-298	P-0034 044-T0 1-IM6		NRP2- ALK	NRP2	in-frame	threep _prime	294485 50	19	1023	True	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		NRP2 (NM_201266) - ALK (NM_004304) rearrangement: c.1291+575_c.3068-119inv Protein Fusion: in frame {NRP2:ALK}	NA
EVT-P-0055 667-T06-IM 7-299	P-0055 667-T0 6-IM7		PICALM- ALK	PICALM	unknown	threep _prime	294463 86	20	1061	False	retained	1.0	False	Protein kinase domain	PF0 062 9; P F12 810		PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) Protein Fusion: mid-exon {PICALM:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0055 667-T02-IM 6-300	P-0055 667-T0 2-IM6		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain	PF00629; P F12810		PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0055 667-T04-IM 6-301	P-0055 667-T0 4-IM6		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain	PF00629; P F12810		PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0054 253-T01-IM 6-302	P-0054 253-T0 1-IM6		PLEKHA7-ALK	PLEKHA7	unknown	three_prime	29448396	19	1035	False	retained	1.0	False	Protein kinase domain	PF00629; P F12810		PLEKHA7 (NM_175058) - ALK (NM_004304) fusion: t(2;11)(p23.2;p15.1)(chr2:g.29448396::chr11:g.16813809) Protein Fusion: mid-exon {PLEKHA7:ALK}	NA
EVT-P-0059 884-T01-IM 7-303	P-0059 884-T0 1-IM7		PPP1CB-ALK	PPP1CB	in-frame	three_prime	29447441	19	1058	True	retained	1.0	False	Protein kinase domain	PF00629; P F12810		PPP1CB (NM_206876) - ALK (NM_004304) fusion: c.52+5708:PPP1CB_c.3172+886:ALKinv Protein Fusion: in frame {PPP1CB:ALK}	NA
EVT-P-0048 893-T01-IM 6-307	P-0048 893-T0 1-IM6		SPTBN1-ALK	SPTBN1	unknown	three_prime	29448346	19	1051	False	retained	1.0	False	Protein kinase domain	PF00629; P F12810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.763+1517:SPTBN1_c.3153:ALKinv Protein Fusion: mid-exon {SPTBN1:ALK}	NA
EVT-P-0053 963-T02-IM 6-308	P-0053 963-T0 2-IM6		SPTBN1-ALK	SPTBN1	unknown	three_prime	29448345	19	1052	False	retained	1.0	False	Protein kinase domain	PF00629; P F12810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-1123:SPTBN1_c.3154:ALKinv Protein Fusion: mid-exon {SPTBN1:ALK}	NA
EVT-P-0045 724-T01-IM 6-309	P-0045 724-T0 1-IM6		SPTBN1-ALK	SPTBN1	in-frame	three_prime	29446796	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; P F12810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-456:SPTBN1_c.3173-402:ALKinv Protein Fusion: in frame {SPTBN1:ALK}	NA
EVT-P-0042 596-T02-IM 6-310	P-0042 596-T0 2-IM6		SQSTM1-ALK	SQSTM1	in-frame	three_prime	29446511	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; P F12810		SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t(2;5)(p23.2;q35.3)(chr2:g.29446511::chr5:g.179254603) Protein Fusion: in frame {SQSTM1:ALK}	NA

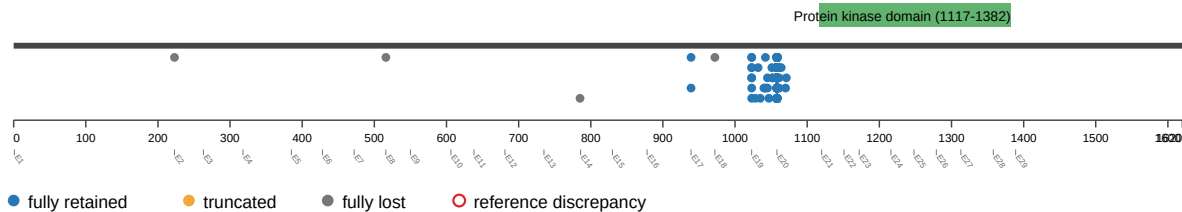
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EVT-P-0042 596-T01-IM 6-311	P-0042 596-T0 1-IM6		SQS TM1- ALK	SQS TM1	in-fr ame	thre e_ _pr ime	294465 11	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t (2;5)(p23.2;q35.3)(chr2:g.2 9446511::chr5:g.17925460 3) Protein Fusion: in frame {SQSTM1:ALK}	NA
EVT-P-0053 901-T01-IM 6-312	P-0053 901-T0 1-IM6		STR N-AL K	STR N	unk now n	thre e_ _pr ime	294463 75	20	1064	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		STRN (NM_003162) - ALK (NM_004304) fusion: c.339 -2565:STRN_c.3192:ALKde l Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0048 102-T03-IM 6-313	P-0048 102-T0 3-IM6		STR N-AL K	STR N	in-fr ame	thre e_ _pr ime	294470 56	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		STRN (NM_003162) - ALK (NM_004304) fusion: c.412 +2761:STRN_c.3173-662: ALKdel Protein Fusion: in frame {STRN:ALK}	NA
EVT-P-0033 066-T01-IM 6-314	P-0033 066-T0 1-IM6		STR N-AL K	STR N	unk now n	thre e_ _pr ime	294483 65	19	1045	False	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		STRN (NM_003162) - ALK (NM_004304) fusion(STRN exons 1-3 fused to ALK exons 19-29): c.413-1 084:STRN_c.3134:ALKdel Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0048 102-T02-IM 6-315	P-0048 102-T0 2-IM6		STR N-AL K	STR N	in-fr ame	thre e_ _pr ime	294470 56	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		STRN (NM_003162) - ALK (NM_004304) fusion: c.412 +2761:STRN_c.3173-662: ALKdel Protein Fusion: in frame {STRN:ALK}	NA
EVT-P-0056 097-T01-IM 6-316	P-0056 097-T0 1-IM6		TFG- ALK	TFG	in-fr ame	thre e_ _pr ime	294478 96	19	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		TFG (NM_001195478) - ALK (NM_004304) rearrangement: t(2;3)(p23. 2;q12.2)(chr2:g.29447896:: chr3:g.100456163) Protein Fusion: in frame {TFG:ALK}	NA
EVT-P-0045 363-T02-IM 6-318	P-0045 363-T0 2-IM6		TPM 1-AL K	TPM 1	in-fr ame	thre e_ _pr ime	294465 14	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		TPM1 (NM_001018020) - ALK (NM_004304) fusion: t (2;15)(p23.2;q22.2)(chr2:g. 29446514::chr15:g.633588 33) Protein Fusion: in frame {TPM1:ALK}	NA
EVT-P-0045 363-T01-IM 6-319	P-0045 363-T0 1-IM6		TPM 1-AL K	TPM 1	in-fr ame	thre e_ _pr ime	294465 14	20	1058	True	retai ned	1.0	False	Protein kinase domai n	PF0 062 9; P F12 810		TPM1 (NM_001018020) - ALK (NM_004304) fusion: t (2;15)(p23.2;q22.2)(chr2:g. 29446514::chr15:g.633588 33) Protein Fusion: in frame {TPM1:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053 383-T01-IM 6-320	P-0053 383-T0 1-IM6		TRAF3-ALK	TRAF3	in-frame	three_prime	29447688	19	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		TRAF3 (NM_003300) - ALK (NM_004304) fusion: t(2;14)(p23.2;q32.32)(chr2:g.29447688::chr14:g.103370282) Protein Fusion: in frame {TRAF3:ALK}	NA
EVT-P-0001 737-T01-IM 3-321	P-0001 737-T0 1-IM3		VCL-ALK	VCL	in-frame	three_prime	29447390	19	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		VCL (NM_014000) - ALK (NM_004304) Translocation : t(10;2)(q22.2;p23.2)(chr10:c.2435-486::chr2:c.3172+937) Protein fusion: in frame (VCL-ALK)	NA
EVT-P-0045 566-T01-IM 6-322	P-0045 566-T0 1-IM6		ZFPM2-ALK	ZFPM2	in-frame	three_prime	29447902	19	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		ZFPM2 (NM_012082) - ALK (NM_004304) fusion: t(2;8)(p23.2;q23.1)(chr2:g.29447902::chr8:g.106384092) Protein Fusion: in frame {ZFPM2:ALK}	NA

Figures

domain retention outliers

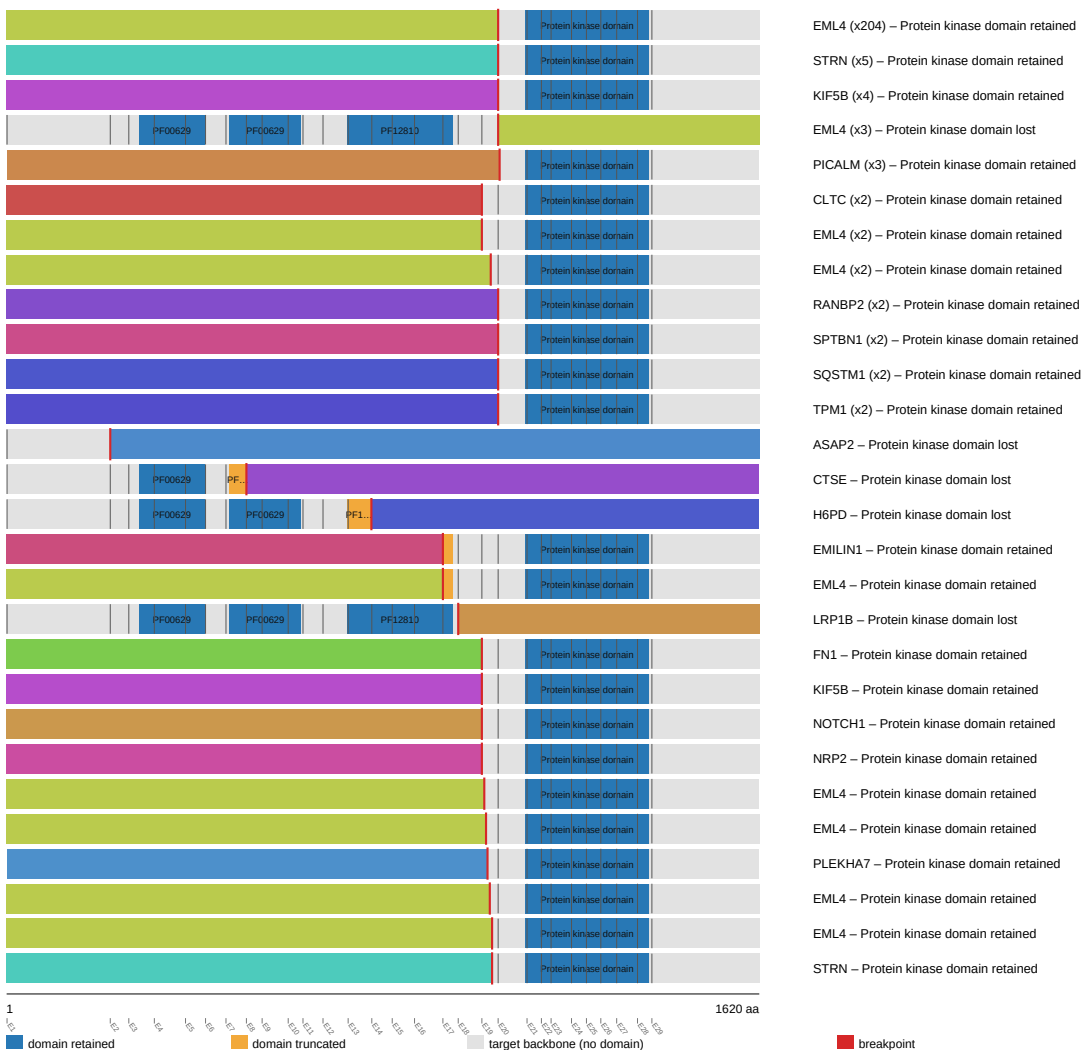
ALK domain-retention track



fusion schematic

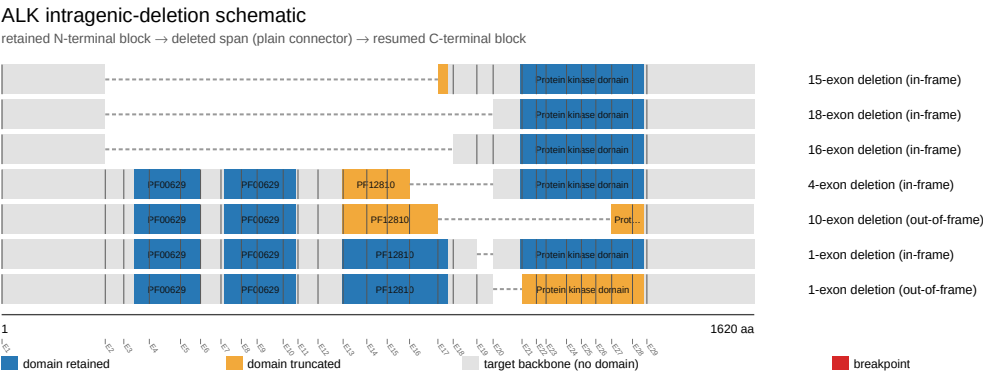
ALK fusion-transcript schematic

partner block → breakpoint → retained ALK portion (domain-colored, exon ticks)



Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

